HAT-MedMNIST

A hybrid agentic training framework for MedMNIST

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Project Manager

scope, schedule, resources, costs, risk management and reporting

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Technical Manager

architecture, agent design, representation and training strategy, integration

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Reviewer

independent review of KPIs, traceability, reproducibility and TRL; the WP6 reviewer agent

Agentic AI / medical-image benchmarks

Outline

1. Motivation, Objectives & Plan
2. System Architecture & MVP
3. Prior Art, Governance & Validation

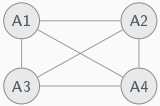
Why HAT-MedMNIST

Script pipeline



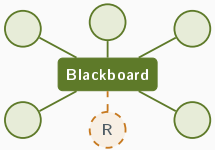
tightly coupled
hard to audit

Typical LLM agents



random at every step
locked to one vendor

HAT-MedMNIST



typed contracts · provenance
independent reviewer with veto

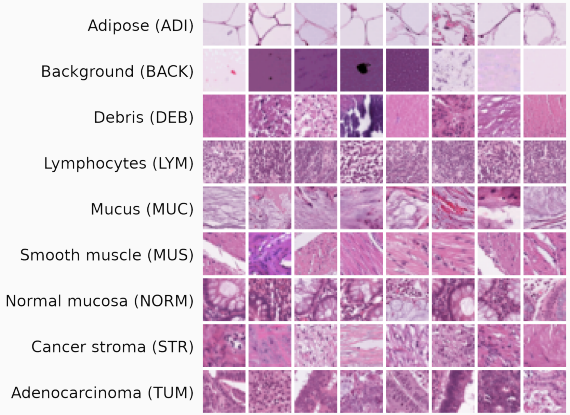
The problem

- Training pipelines are monolithic scripts: hard to audit and to reproduce.
- LLM-agent frameworks let a model improvise every step, tied to one vendor.

Our answer

- One duty per agent; agents exchange only versioned artefacts.
- LLM reasoning only at declared decision points, with a fallback.
- An independent reviewer audits every stage and can veto.

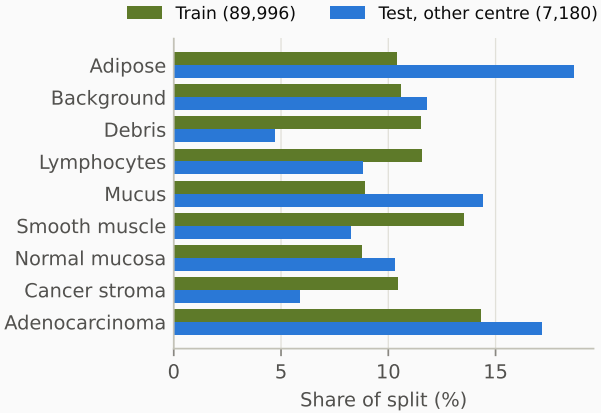
The task in pictures: PathMNIST



Can software agents learn to tell colon tissue types apart?

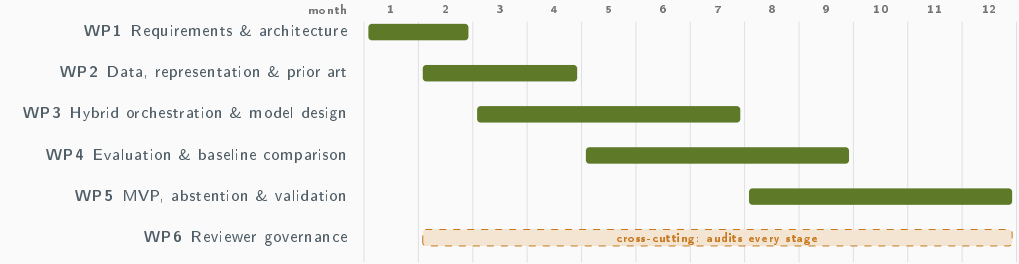
- PathMNIST: stained (H&E) patches from colorectal cancer slides, part of MedMNIST v2.
- 9 tissue classes, 107,180 images of only **28×28 pixels**.
- Train/validation from one source; test from **a different hospital**.
- Small images keep experiments fast; the second hospital keeps the test honest.

The data at a glance



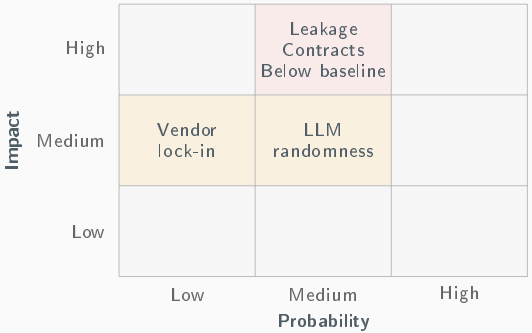
- Official split: 89,996 / 10,004 / 7,180 images.
- Training classes are near-balanced (largest/smallest = 1.63).
- The test hospital differs: adipose 18.6% vs 10.4%, debris 4.7% vs 11.5%.
- So test accuracy measures **transfer** to new data, not memory.

The plan: six work packages in twelve months



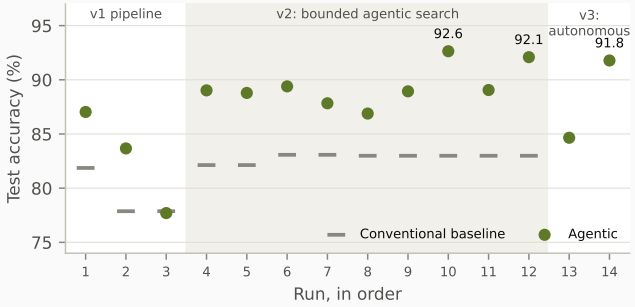
- Hybrid Agile / stage-gate: short iterations, a formal gate at the end of each WP.
- Each WP delivers a reviewable output (D1–D6) with numeric KPIs.
- WP6, the reviewer, runs alongside every other WP.
- Goal: an integrated MVP demonstrated at **TRL 7** in month 12.

Risk map



- **Leakage**: an audit gate blocks training until splits and statistics are checked.
- **Contracts**: typed artefacts fail fast; the orchestrator can veto.
- **Below baseline**: baseline-first, ablations, negative results reported.
- **LLM randomness**: temperature 0, logged provenance, fallback.
- **Lock-in**: any open model behind one interface.

From first run to MVP: 14 runs, three versions



- **v1**: first end-to-end runs; one tie with the baseline.
- **v2**: agents choose within a bounded search; every run beats the baseline, best 92.6%.
- **v3**: fully autonomous; agents write and launch their own experiment code.
- Same official split throughout, so the runs are comparable.

The flagship run in numbers

3 h 31 min
end-to-end, full dataset

8
model trials, chosen by agents

28
LLM decisions, local model

72
versioned artefacts

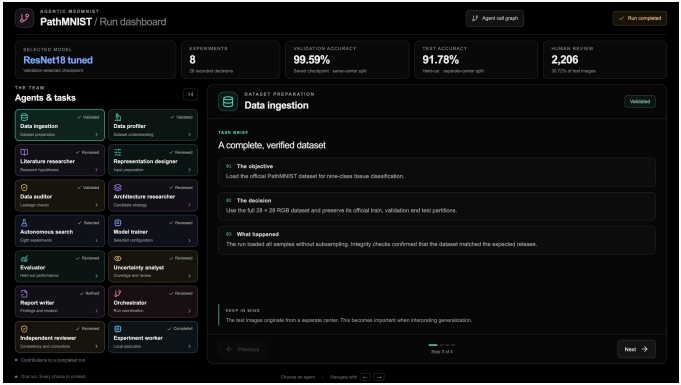
What ran

- 11 stages on the full dataset, one local CUDA machine.
- LLM: qwen3.8 served locally by Ollama; no cloud model.
- The search agent proposed and launched all 8 trials.

Where the time went

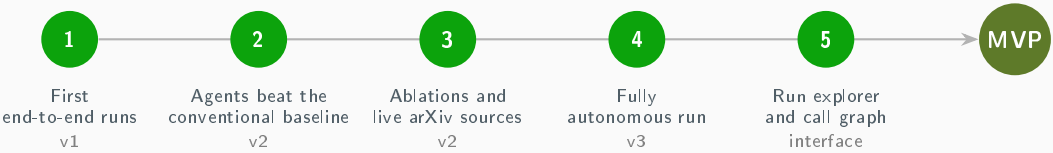
- Model search: 3 h 20 min, 95% of the run.
- All other stages together: about 11 minutes.
- 56 recorded calls: 12 stage starts, 12 reviews, 32 worker jobs.

Run explorer: the run at a glance



- One page per run: chosen model, experiments, validation and test accuracy, human-review load.
- 14 agent cards; each opens the objective, the decision and what happened.
- Built for non-specialists: no code, prompts or raw files on screen.
- Opens locally from the stored results; no training or LLM needed.

Milestones reached



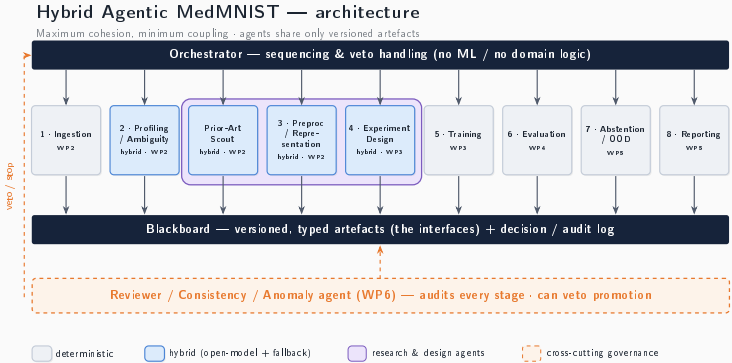
Delivered

- 11-stage agentic pipeline with an independent reviewer on every stage.
- Autonomous model search that writes and runs its own experiments.
- Run explorer that shows every decision and call.

Key numbers

- **92.1%** vs 83.0% for the conventional pipeline (v2).
- **91.8%** from the fully autonomous run (v3).
- **13** acceptance criteria evidenced across the runs.

The Blackboard architecture



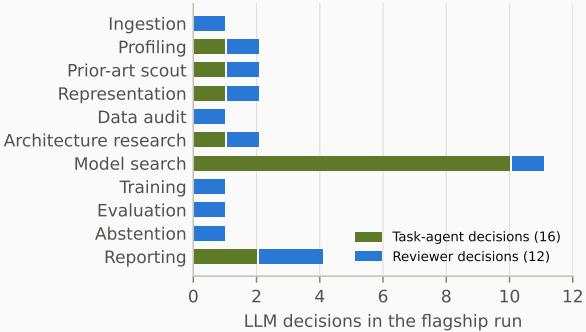
Coordination

- **Orchestrator**: sequences stages; holds no domain logic.
- **Blackboard**: versioned, typed, checksummed artefacts plus a decision log.

Control

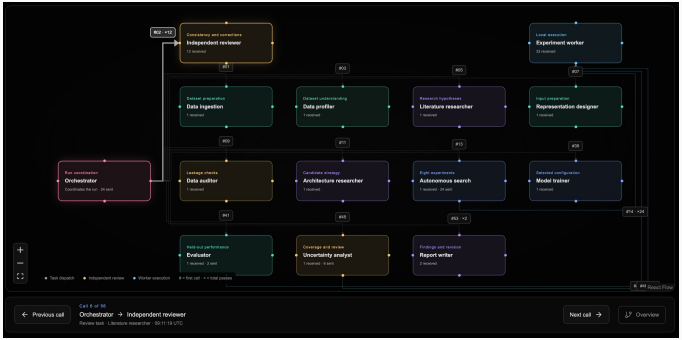
- **Reviewer**: audits after every stage; can request a retry or veto.
- **0 direct agent calls**, verified on the code's syntax tree.

Where the LLM is allowed to decide



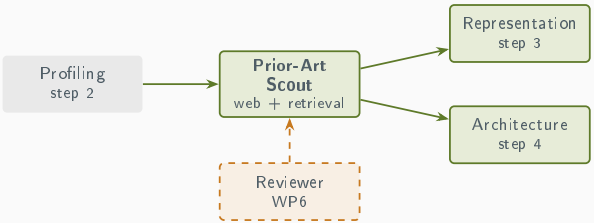
- **Deterministic**: ingestion, data audit, training, evaluation, abstention. No task-agent LLM call.
- **Judgement**: ambiguity, prior art, representation, architecture, search, report.
- Model search uses most decisions: 10 of 16.
- Every decision is logged with its prompt and source; the heuristic fallback was used once.

Run explorer: agent call graph



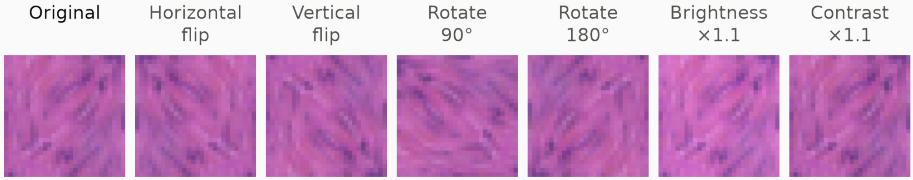
- One box per agent; 56 recorded calls grouped into 15 connections.
- Calls go through the orchestrator, the reviewer or the worker: **no agent calls another**.
- Step through calls in time order, with timestamps and task context.
- 12 stage dispatches, 12 reviews, 32 worker jobs.

The Prior-Art Scout



- Searches arXiv; every source is snapshotted and checksummed.
- Writes a cited brief that feeds representation and architecture.
- Ranks existing options, or proposes new ones that wait at a review gate.
- Advises only: never trains, selects or calls another agent.
- Live sources in the v2 run: 6 of 6.

Representation: what the model sees



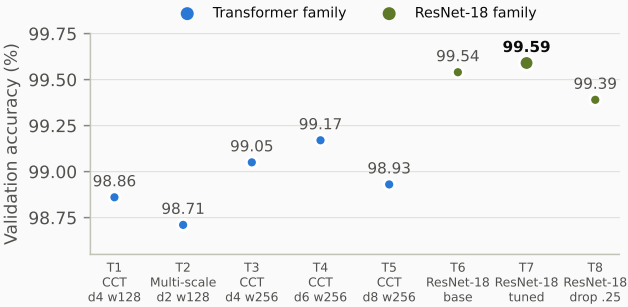
Normalisation

- Per-channel standardisation.
- Statistics from **training images only**, reused unchanged for validation and test.

Augmentation

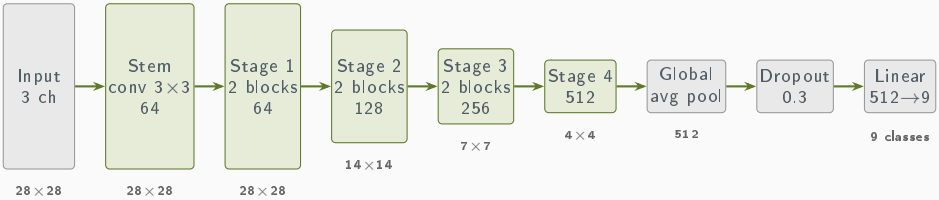
- 6 transforms picked by the representation agent from a label-preserving whitelist.
- Tissue has no fixed orientation, so flips and rotations keep the label.
- 89,996 images $\rightarrow$ 629,972 training views.

Autonomous model search



- The agent reads every past result before choosing the next trial.
- Trials 1–5: compact and multi-scale transformers, best 99.17%.
- Trial 6: switch to ResNet-18, immediately better (99.54%).
- Selected on validation accuracy, macro-F1 as tie-break. **Test never seen.**

The winning architecture: ResNet-18 for 28×28



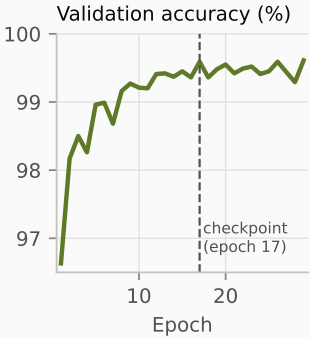
Network

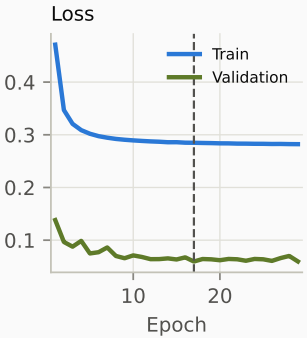
- CIFAR-style ResNet-18: 3×3 stem and **no max-pool**, so small images keep detail.
- 4 stages $\times$ 2 residual blocks, widths $64 \rightarrow 512$.
- **11.17 M** parameters, trained from scratch.

Training recipe

- AdamW, lr 2×10^{-4}, weight decay 0.01, cosine schedule.
- Class-weighted loss, label smoothing 0.05, dropout 0.3.
- Batch 256; early stopping with patience 12.

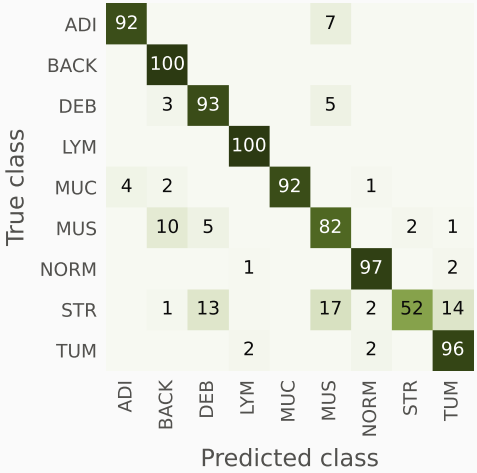
Training dynamics





- Validation accuracy passes 99% by epoch 5.
- Best checkpoint: epoch 17, **99.59%**.
- Early stopping at epoch 29, after 12 epochs without improvement.
- Train loss stays near 0.28 because of label smoothing and augmentation, not underfitting.

MVP result on the held-out hospital



91.8% test accuracy

0.981 macro AUC

0.890 macro-F1

- 7,180 images from a hospital never seen in training; 95% CI 91.1–92.4%.
- 7.8 points below validation: the cost of changing hospital.
- Lymphocytes and background are perfect; cancer stroma is the weakest row.

Leakage-safe by design



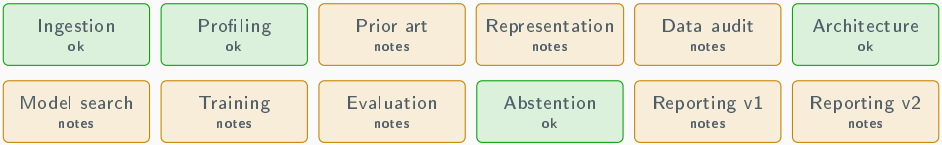
Rules

- Normalisation fitted on training images only.
- Model selection on validation only.
- Test used once, after the model is frozen.

Evidence in the run

- **11/11** audit checks passed before training.
- **0** images shared between train and validation.
- **0** test metrics used in selection.

The reviewer at work



ok = no issues notes = minor issues logged for the record

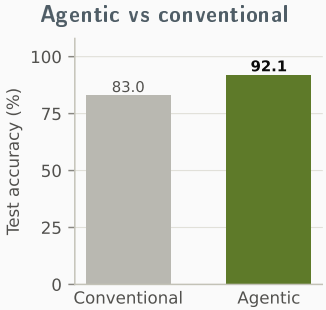
What it checks

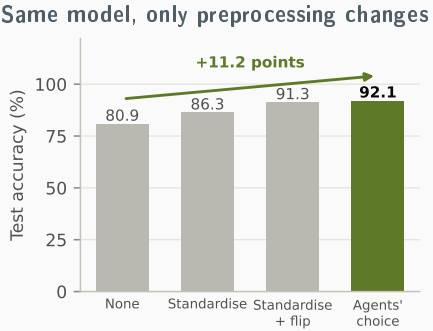
- Consistency and arithmetic of every artefact.
- Leakage, provenance and citation validity.
- Written claims against the actual numbers.

What it did

- 12 reviews: 4 clean, 8 with minor notes, **0 vetoes**.
- Caught a rationale claiming 16 image tokens instead of 49.
- Requested a rewrite of the final report (v1 → v2).

Agents vs a conventional pipeline

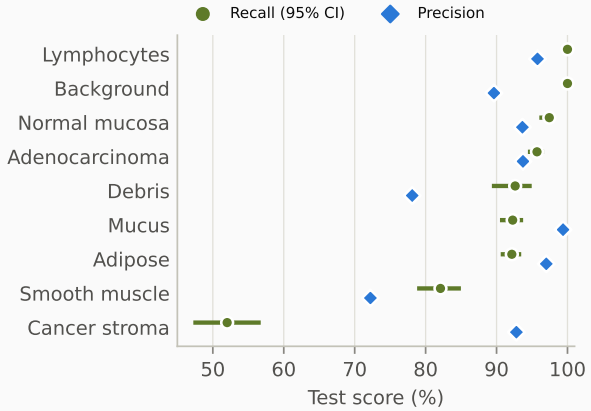




Preprocessing drives the result

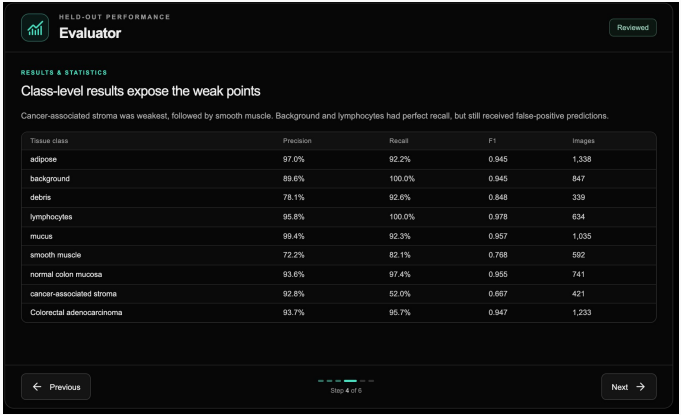
Same model, same training: the agents' preprocessing (standardisation, flip, brightness, contrast) alone lifts accuracy from 80.9% to **92.1%** (**+11.2 points**). Overall, agents beat the fixed recipe by +9.1 points.

Error analysis: which tissues fail, and why



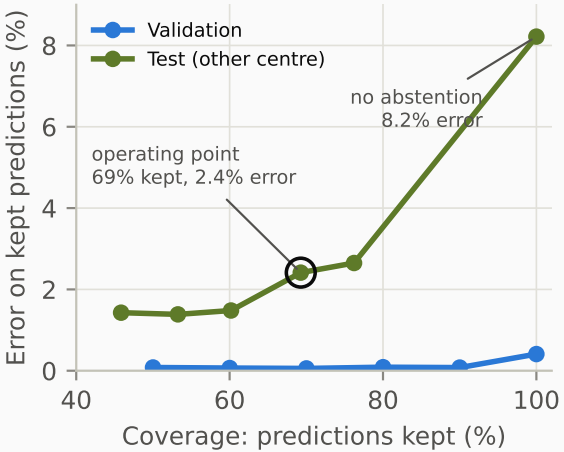
- 7 of 9 tissues reach $\geq$92% recall at the new hospital.
- **Cancer stroma**: 52% recall but 93% precision. Missed cases go to muscle (17%), adenocarcinoma (14%), debris (13%).
- At 28 pixels these tissues share fibrous texture.
- The profiling agent flagged this risk **before training**.

Run explorer: auditing the evaluation



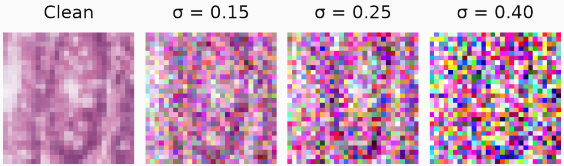
- Per-class precision, recall and F1, exactly as the evaluator agent computed them.
- Every statistic is read from the versioned artefacts, so anyone can re-check it.
- The weak class, cancer stroma, sits next to the headline accuracy, not hidden by it.
- Same numbers as in this talk: one source of truth.

Knowing when not to answer



- The model abstains when its predictive entropy is high.
- Threshold set on validation for 80% coverage; never tuned on test.
- Test: keeps 69% at **97.6%** accuracy; 2,206 images go to human review.
- Coverage drops at the new hospital: the model knows it is less sure.

Robustness check: corrupted inputs



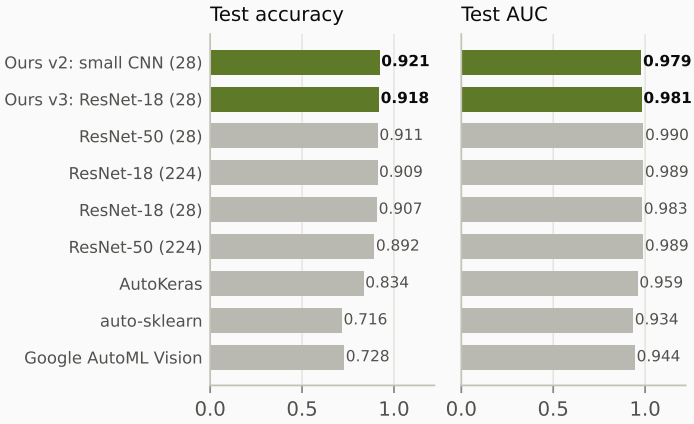
Test

- Gaussian pixel noise at three strengths.
- Question: does the abstention score separate noisy from clean images?

Result

- AUROC **0.943** on test (0.994 on validation).
- At most 0.2% of noisy images accepted.
- Limit: noise is a proxy; staining and scanner shifts are untested.

Comparison with the state of the art



Published: MedMNIST v2, Yang et al., *Sci. Data* 2023, Table 3 (mean of $\geq$3 runs).

Accuracy: **above every published baseline**.

AUC: slightly below the best ResNets.

Far ahead of AutoML tools, the closest kind of automated pipeline.

Acceptance evidence

Criterion	Status
Typed, checksummed hand-offs	✓
Zero direct agent calls	✓
Leakage audit before training	✓
Cited ideas	✓
Live web retrieval	✓
Validation-only selection	✓
Baseline comparison	✓

Criterion	Status
Representation ablations	✓
Run-to-run variance	✓
Abstention and human hand-off	✓
Reviewer report on every stage	✓
Decision provenance	✓
End-to-end run	✓

13

acceptance criteria evidenced
in the project runs

Thank you

Maximum cohesion, minimum coupling.